MILITARY SPECIFICATION

SIGNAL DATA CONVERTER MK 72 SERIES

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope. - This purchase description covers the requirements for Signal Data Converter Mk 72 which will serve as an interface between the digital computer and radar set which comprise the main elements of Guided Missile Fire Control Systems Mk 74 Mods 4, 5, 6, or 7 as herein defined.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on data of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FSC 1220

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SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-S-901	Shock Test, H. I. (High-Impact), Shipboard Machinery, Equipment and Systems, Requirements for
MIL-F-18870	Fire Control Equipment, Naval Ship and Shore, General Specification

STANDARDS

Federal

FED-STD-151A	Metal, Test Methods

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment
MIL-STD-210	Climatic Extremes for Military Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Demonstration
MIL-STD-740	Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment

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STANDARDS (cont.)

Military

MIL-STD-781	Test Levels and Accept/Reject Criteria for Reliability of Non-Expendable Equipment

PUBLICATIONS

Bureau of Naval Weapons

QAP 2868644	Quality Assurance Provisions for Signal Data Converter Mk 72 Mod 0
QAP 2884550	Quality Assurance Provisions for Signal Data Converter Mk 72 Mod 1

Military

MIL-HDBK-217	Reliability Stress and Failure Rate Data for Electronic Equipment
MIL-HDBK-472	Maintainability Prediction

DRAWINGS

Naval Ordnance Systems Command

IL 2868644	Signal Data Converter Mk 72 Series

(Copies of Specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Functional description. - Signal Data Converter Mk 72 series, which may hereinafter be referred to simply as the SDC, shall perform data conversions necessary for Digital Computer Mk 152 Mod 0 or 1 to operate with Guided Missile Fire Control System (GMFCS) Mk 74 Mods 4, 5, 6, or 7 and other elements of the TARTAR Weapon System.

Equipments of the fire control system and elements of the weapons system may hereinafter be referred to collectively (with respect to the SDC) as simply as the system.

The major functions of the SDC shall be as follows:

a) Convert linear ac and dc and synchro signals to digital form.
b) Convert digital signals to linear ac and dc and synchro.
c) Transfer digital signals (logic and relay) between computers and peripheral units.
d) Communicate in time-multiplex fashion, with two computers.
e) Perform self-test functions manually and automatically under control of a computer test program.

3.1.1 Analog-to-digital (A/D) conversions. - A/D conversion is the transformation of the ratio of an analog signal and its reference to a binary (digital) number. The types of analog signals which shall be converted to digital form in the SDC are summarized in Table I.

3.1.1.1 Static A/D accuracy measured over the full dynamic range. -

Channel Type and Accuracy	% of Samples Exceeding Stated Accuracy
ac and Type 2 dc linear A/D*	
12 bits ± 1 LSB	1.0%
11 bits ± 1 LSB	0.5%
10 bits ± 1 LSB	0.1%
9 bits ± 1 LSB	0.01%

* Types 1 and 2 dc linear signals are determined by the respective accuracy requirements set forth in Table I

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TABLE I. - Analog-to-digital conversion specifications

Conversion Type	Conversion Time (max) (1)	Accuracy (Nominal)	Z_{in} (min)	Voltage Level Full Scale	Ground Isolation (min) (3)
dc linear (Type 1)	150 µs	±1.25%	100 kΩ	±20, ±40, ±80	100 kΩ
dc linear (Type 2)	150 µs	±0.05% (12 bits)	100 kΩ	±20, ±40, ±80	100 kΩ
ac linear (400 Hz)	150 µs	±0.05% (12 bits)	100 kΩ	±12 V rms (4) ±24 V rms	100 kΩ
Single-speed synchro	150 µs	±5.3 (min) (12 bits)	100 kΩ	90 V rms line-to-line	10 MΩ
dual-speed synchro	200 µs	±10 (min) (2) (17 bits)	100 kΩ	90 V rms line-to-line	10 MΩ

NOTES:

(1) Conversion time includes computer control time
(2) ±10 min accuracy assumes a synchro speed ratio of 32:1 or greater
(3) For frequencies less than 1 kHz
(4) ± indicates that the converter is phase sensitive

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Channel Type and Accuracy	% of Samples Exceeding Stated Accuracy
Type 1 dc linear A/D*	
10 bits ± 1 LSB	0.5%
9 bits ± 1 LSB	0.1%
8 bits ± 1 LSB	0.01%
1X S/D	
12 bits ± 1 LSB	1.0%
11 bits ± 1 LSB	0.5%
10 bits ± 1 LSB	0.1%
9 bits ± 1 LSB	0.01%
1-36X S/D	
17 bits ± 1 LSB	1.0%
16 bits ± 1 LSB	0.5%
15 bits ± 1 LSB	0.1%
14 bits ± 1 LSB	0.01%
2-36X S/D	
16 bits ± 1 LSB	1.0%
15 bits ± 1 LSB	0.5%
14 bits ± 1 LSB	0.1%
13 bits ± 1 LSB	0.01%

* Types 1 and 2 dc linear signals are determined by the respective accuracy requirements set forth in Table I.

3.1.1.2 Dynamic A/D accuracy. - The nominal break frequencies of the noise filters in the input analog signal conditioners shall not be less than those shown below.

a) dc linear - The frequency response can be approximated by a single break (6 decibels/octave) with a 3 decibel point of 1 kilohertz minimum.
b) ac linear - The frequency response can be approximated by a double break (12 decibels/octave) with a 6 decibel point of 40 hertz.
c) 1X synchro - The frequency response can be approximated by a single break (6 decibels/octave) with a 3 decibel point of 40 hertz.
(produces 5 seconds of error at 3.5 rpm)
(and 30 seconds of error at 21 rpm)

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3.1.2 Digital-to-analog (D/A) conversions. - D/A conversion is the transformation of a binary (digital) number to an analog voltage proportional to its reference voltage. The types of digital signals which shall be converted to analog form are summarized in Table II.

3.1.2.1 Dynamic D/A transients. -

a) ac and Type 2 dc linear - Settling time to stated accuracy after a step change in the output value shall be less than 2 milliseconds. Switching transients on slowly-varying outputs shall have an amplitude less than 10 percent of full scale and a duration of less than 100 microseconds.
b) Type 1 dc linear (sample and hold) - Settling time to stated accuracy after a step change in the output value shall occur after less than 10 output commands of the new value. Switching transients - same as Type 2.
c) Synchro - Settling time to stated accuracy after a large (greater than 5 degree) step change in output value shall be less than 100 milliseconds. Undesirable transients for small (less than 5 minutes) changes in output value shall be less than 30 seconds and shall have a duration of less than 5 milliseconds.

3.1.3 Digital data transfer. -

3.1.3.1 Digital output channels (DOC's). - The SDC shall transmit digital data (30-bit words) received from a controlling computer to each DOC. The standard Navy slow interface shall exist between the device and each DOC. The SDC shall be able to address eight peripheral devices on each DOC trunk with a maximum capability of three (3) DOC trunks.

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TABLE II. - Digital to analog conversion specification

Conversion Type	Resolution	Accuracy	Conversion Time (max) (1)	Maximum Voltage Level	Maximum Load (5)	Ground Isolation (4)
dc linear (Type 1)	10 bits	±0.5% (7)		±40 V, ±80 V ±20 V	10 kΩ 10 kΩ 500 Ω	100 kΩ
dc linear (Type 2)	12 bits	±0.1%	30 µs	±40 V, ±80 V ±20 V	10 kΩ 10 kΩ 500 Ω	100 kΩ
ac linear (400 Hz)	12 bits	±0.1%	30 µs	±12 V, ±24 V ±12 V	10 kΩ 10 kΩ 500 Ω	100 kΩ
Single-speed synchro (3)	12 bits	±10.5 minutes (6) (11 bits)	30 µs	90 Vrms line-to-line 11.8 Vrms line-to-line	2500 Ω line-to-line 300 Ω line-to-line	10 MΩ
Dual-speed synchro (3)	17 bits	±19.8 seconds (2) (16 bits)	50 µs	90 Vrms line-to line	2500 Ω line-to-line on each speed	10 MΩ

NOTES:
(1) Conversion time includes computer control time
(2) With a speed ratio of 32:1 or greater; full accuracy requires the load synchros to be at null
(3) Synchro signals are 400 Hz carrier
(4) For frequencies less than 1 kHz
(5) Max load is determined by power saturation characteristics of amplifier
(6) Full accuracy requires the load synchros to be at null
(7) With external reference

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3.1.3.2 Demand-digital channels (DDC's) and demand digital interrupt channels (DDIC's). - The SDC shall accept demand digital data (30-bit words) upon computer command and transfer the data to the controlling computer. The standard Navy slow interface shall exist between the device and each DDC. Each DD channel shall be capable of handling 8 peripheral devices. Four channels may be converted to a DDIC independently to allow peripherals to transfer high priority data with minimal delays via the interrupt facility.

3.1.4 Relay logic transfer. -

3.1.4.1 Logic input channels (LIC's). - The SDC shall transfer 30-bit status relay level data to a controlling computer via each LIC. Relay voltages can be +28 volts dc, +50 volts dc and 115 volts rms, 60 hertz and 400 hertz. The control voltage returns shall not be grounded within the SDC. Current requirements are:

a) 25 milliamperes dc nominal
b) 30 milliamperes ac rms
c) 75 milliamperes ac peak

3.1.4.2 Logic output channels (LOC's). - The SDC shall transfer 30-bit control digital data from the controlling computer to LOC relays for contact closures for use by external control systems. The contacts shall be capable of handling 50 volts dc at 0.5 amperes or 115 volts ac 60/400 hertz at 1 ampere.

3.1.5 Modes of operation. - The SDC shall have two primary modes of operation: normal and test.

3.1.5.1 Normal mode. - The SDC shall normally operate under computer control unless placed in a manual test mode.

3.1.5.2 Test mode. - The SDC shall have test features that permit a rapid determination of the operational readiness of the equipment. In the event that a malfunction is detected, additional test features shall provide an expeditious means of locating the faulty plug-in assembly.

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3.1.5.2.1 Computer aided test. - In this mode, a computer test program will automatically command end-around (rebound) tests of similar input and output channels, compare the transmitted signal against the received signal, and output errors revealed by the comparison. The SDC shall be able to perform automatic rebound functions without the need of external switching and simultaneously operate in the normal mode on those channels not selected for rebound tests.

This will be done by incorporating a method for rapidly testing those circuits within the input and output sections of the SDC for which it is possible to have the outputs of the circuits in the output section provide the inputs to circuits of the same type in the input section. It shall be possible to test the operation of the ac, dc, synchro, logical, and digital input and output circuits. The inputs and outputs shall be connected to each other through internal switching to a common bus line for each data type within the SDC. Only one output/input combination, selected either manually or under computer control, shall be required to be tested at one time. Under computer control, a bit located in the "c" bits of format 1 shall determine whether the selected channel is to be disconnected from the system for "end-around" tests. When the SDC is under computer control, it shall operate in the normal mode of operation.

NOTE - It is to be noted that since there is no 11.8 volt line-to-line synchro input capability, the 11.8 volt line-to-line synchro output will be rebound tested against the 90 volt line-to-line input channel which will result in an accuracy degradation during test of ±1 degree.

3.1.5.2.2 Manual self-test mode. - In this mode, the computer does not control the tests. Channel numbers and data for the rebound tests are selected manually at the SDC.

3.1.5.2.3 Local test mode. - In this mode, each channel is checked manually for performance using test equipment for both source voltages and measuring devices.

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3.2 Performance requirements. -

3.2.1 Computer-SDC interface. - The SDC shall operate as specified herein with the computers listed below:

a) CP-642A (NAVSHIPS 0967-051-6020)
b) CP-642B (NAVSHIPS 0967-051-6040)
c) UNIVAC 1230
d) CP-789*
e) Digital Computer Mk 152 Mod 1 (WS 11516)*
f) Digital Computer Mk 152 Mod 0 (WS 7891)*
g) Digital Data Computer AN/UYK-7*

*Operating in the dual-channel configuration

For the purposes of this specification in referencing computer-SDC interface, the term "input" shall refer to input from the SDC and the term "output" shall refer to output from the computer to the SDC, unless otherwise indicated.

The conversion equipment shall use the External Function (EF), Output Data Request (ODR), and Output Acknowledge (OA) control lines on output channels and shall use the Interrupt (INT), Input Data Request (IDR), and Input Acknowledge (IA) control lines of input channels.

3.2.1.1 Duplexing operation. - The SDC shall be capable of communicating with 2 computers in a time-multiplexed manner. An ODR shall be raised to both computers at turn on. The SDC will interrogate computer A and computer B OA lines alternately. The computer whose set OA line is detected first by the SDC will obtain initial control of the SDC.

If a computer requests control (i.e., raises the OA line) while the SDC is busy that computer's request shall be remembered and honored immediately upon the initiation of the next interrogation cycle.

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The SDC will inhibit communications with a computer if (1) a computer does not respond to an ODR within 200 microseconds when additional data is required to complete a command or (2) a computer does not respond to an IDR within 200 microseconds. However, any computer after losing communications, may be reinstated by raising its External Function (EF) line to the Conversion Equipment for a minimum of 4 microseconds, constituting a Master Clear operation (MSTC).

3.2.1.2 SDC command structure. - Communication between the SDC and a computer are conducted using the computer word formats shown in Figure 1.

Format 1 is used for all commands except DOC to the SDC. Format 1A is used for DOC commands. Format 2 is used for transferring digital data between the SDC and the computer.

Format 3 is used to transmit mode commands to the SDC control structure.

Properly established computer Input/Output buffers shall always be satisfied by the SDC regardless of the addressed peripheral units.

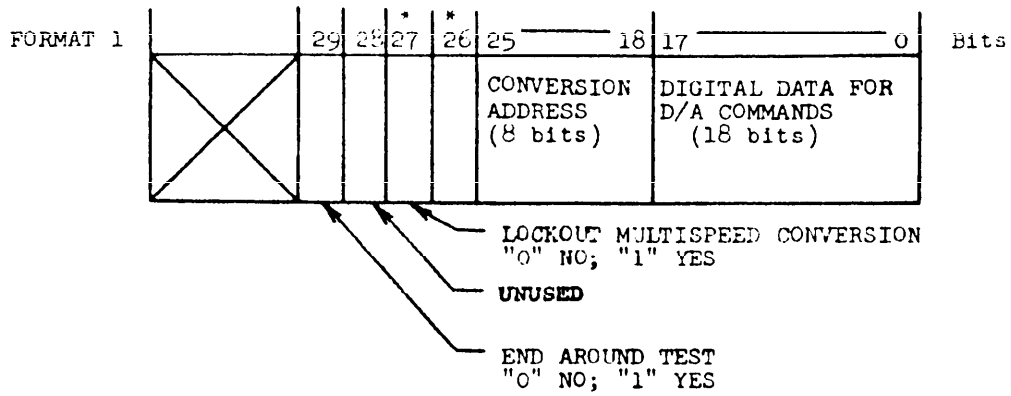
3.2.1.3 Interrupt input transfers. - A computer shall receive Interrupt data from a peripheral unit connected to a DDIC. The SDC always raises the Interrupt line to the computer assigned to the DDIC (Computer/DDIC assignments will be determined later). The computer responds by (1) accepting the Interrupt data on the Input Data Lines and (2) acknowledging receipt of data by sending an IA signal to the SDC. The Interrupt data format is shown in Figure 2.

3.2.2 Peripheral/SDC/interface. -

3.2.2.1 Demand Digital (DD) interface. - (Figure 3)

3.2.2.1.1 Normal mode. - The SDC shall be capable of communicating with up to eight (8) Demand Digital devices, (a) with each device assigned to a DD address. The input from the DD Deivce shall be of Format 2 configuration (see Figure 1). The computer shall use a Format 1 command to

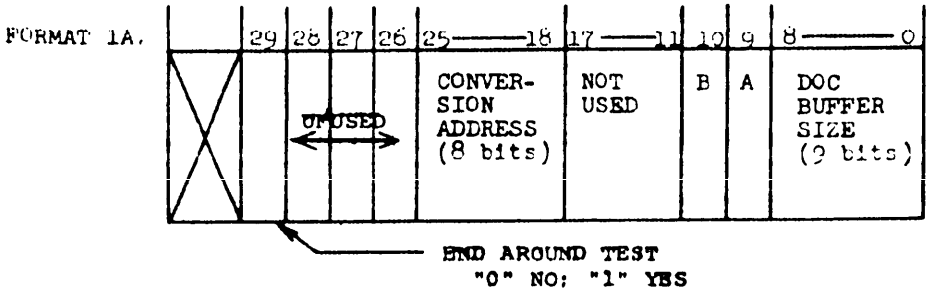
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*NOTE - Bit 26 and 27 are mode dependent
In Synchro Mode - Bit definitions are as indicated in chart
In Linear Mode - Bit 26 and 27 are used to select a reference

BIT 26	BIT 27	
0	0	Internal Reference
0	1	Reference A
1	0	Reference B
1	1	UNUSED

These codes will apply to blocks of A/D and D/A channels if more than three (3) external references are required.



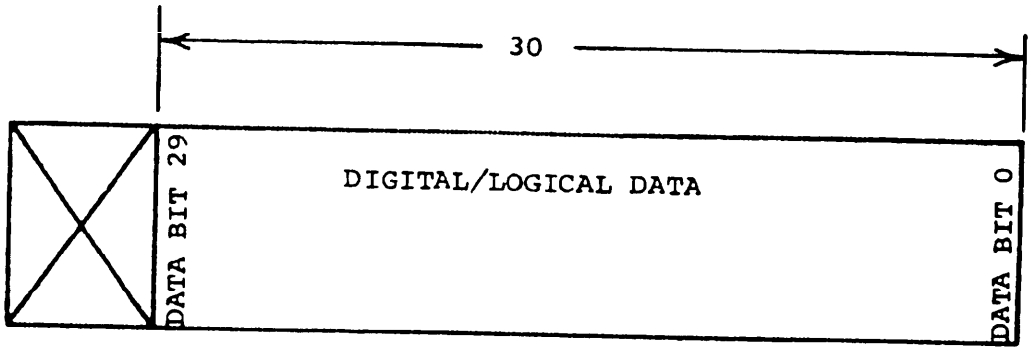
A - BIT 9 (a) All data words will be sent out with ODR/OA control lines
(b) There will be no-automatic Master Clear (MSTC) and/or Receive Data Unit Computer RDUC
State "1" Affirms (a) and (b)
State "0" Denies (a) and (b)

B - BIT 10 Any data coming with this command will be forced (with the generation of a pulse) on the external function line.

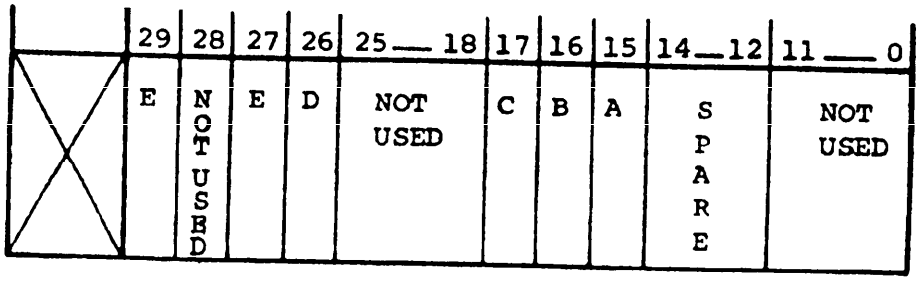
Figure 1. - Computer-SDC word formats

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FORMAT 2



FORMAT 3



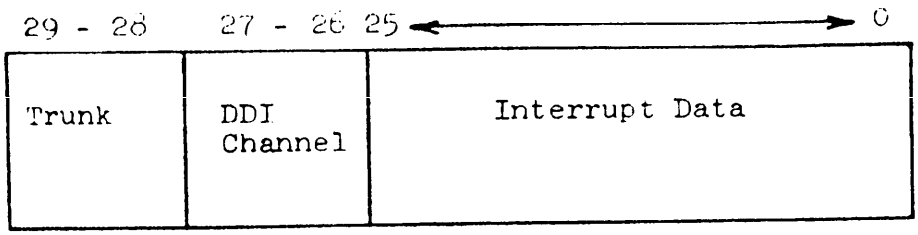
A - BIT 15 State "1" indicates Enable Interrupt from Demand Digital Trunk #1
B - BIT 16 Same as Bit 15 for DD Trunk #2
C - BIT 17 Same as Bit 15 for DD Trunk #3
D - BIT 26 Master Clear (MSTC) Clear = "1"
No Clear = "0"
E - BIT 27 Inhibit normal multispeed state = "1"
Normal multispeed = "0"

E - BIT 29 End Around test = 1
EAT-MSTC = 0

NOTE: All above functions stay in state until changed by External Function

Figure 1. (continued)
Computer-SDC word formats

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Bits 0 - 25 (26 bits)
Interrupt Data

Bits 26 - 27 (2 bits)
DDI Channel in Trunk

Bit 26	27	
0	0	1st DDI address
1	0	2nd DDI address
0	1	3rd DDI address
1	1	4th DDI address

Bits 28 - 29 (2 bits)
Trunk

Bit 28	29	
0	0	Trunk No. 1
1	0	Trunk No. 2
0	1	Trunk No. 3

Figure 2 - DDIC Interrupt Data Word

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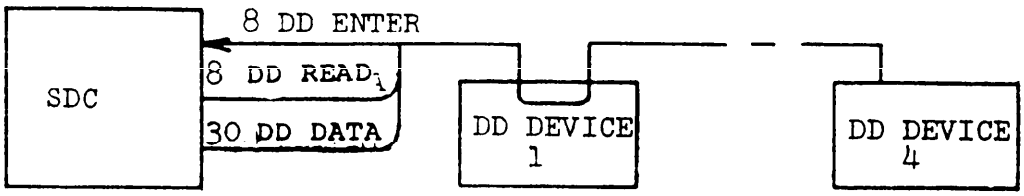


Figure 3. Demand Digital Interface

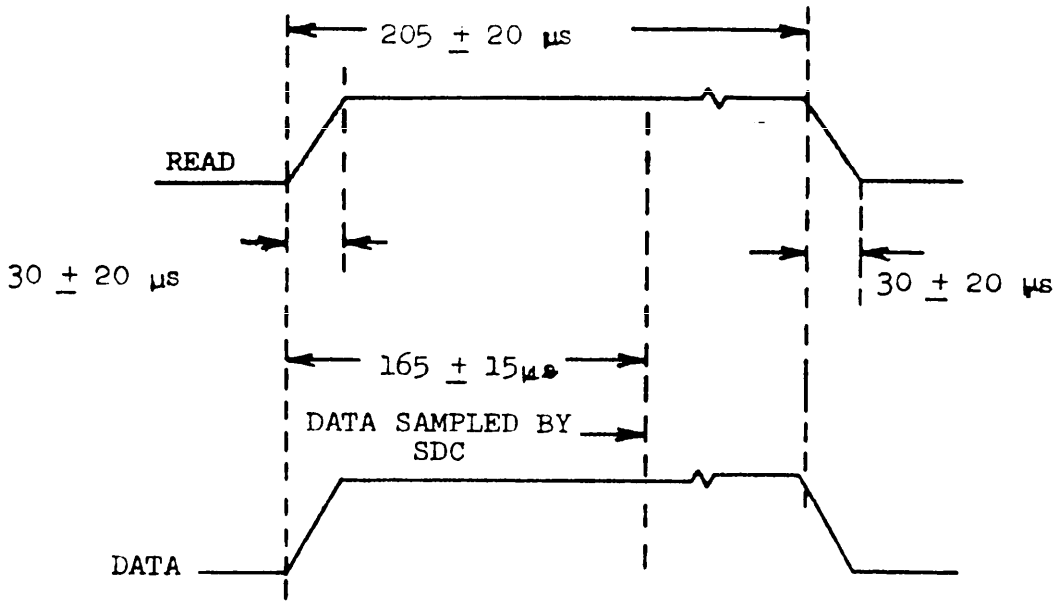


Figure 4. Demand Digital Timing

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select the DD Device of interest. The SDC shall then monitor the addressed DD "Enter" line for a period of 220 ± 20 microseconds. If during this interval the "Enter" signal is detected, a "Read" signal is used to transfer the stored information from the DD Device to the SDC. If the "Enter" signal is not received, the SDC shall set the input data to the controlling computer to all "ones" indicating a lack of data. Actual sampling of the DD Data shall occur at a time of 165 ± 15 microseconds after initiation of the "Read" signal, to allow for transient setting (see Figure 4).

3.2.2.1.2 Interrupt mode. - If a DD Interrupt device wishes to send interrupt data to the computer, it raises it's "Enter" line. If the DD Channel is in the Interrupt Mode, a transition from "0" to "1" of the "Enter" line associated with that channel will be interrupted as an interrupt by the DD Device on that channel. If the DD Interrupt occured while the SDC was performing a conversion or data transfer, the SDC shall complete that data processing (and transfer data to computer -- if required) and then raise the "Read" line according to a prescribed priority. The SDC shall then transfer the interrupt data of the highest priority DDID channel into the computer "input holding register". The highest priority on the DDI Channels shall be from the first interrupting channel of the first trunk sequentially through all interrupting channels to the last interrupting channel of the third trunk. After the interrupt data has been placed in the computer "input holding register", the SDC shall raise the interrupt line to a computer initiating transfer of the data.

3.2.2.1.2.1 Interrupt during conversion or transfer. - The computer to which the SDC shall send the interrupt data shall be the computer involved in the interrupted conversion or transfer unless the computer has locked out interrupts from that DD Interrupt Channel; in which case the other computer will receive the interrupt, unless it has the locked-out channel also, in this case, the interrupt will be lost.

3.2.2.1.2.2 Interrupt during duplexing. - During duplexing the computer being interrogated at the time the interrupt is received, will become the computer defined in the Interrupt mode description as being the computer whose conversion or transfer was interrupted.

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3.2.2.1.2.3 Multiple interrupts. - The Interrupt mode shall be repeated until all interrupts have been honored. The computer shall acknowledge receipt of all interrupt data by sending an Input Acknowledge (IA) signal to the SDC, for each interrupt.

3.2.2.2 Digital output channel (DOC) interface. - The DOC interface circuits shall provide the interface necessary for transferring data from the SDC to the peripheral units attached to one of the DOC's (Figure 5).

The DOC interface in the SDC shall have the following control lines; External Function (EF), Output Data Request (ODR), and Output Acknowledge (OA). Data may be forced out of a DOC using an External Function, data may be requested from the DOC by an Output Data Request answered by the DOC with an Output Acknowledge, or a combination of both may be used.

(NOTE: The most common method of data transmission on the DOC consists of an External Function forcing out a Master Clear (MSTC) to all the peripherals on a DOC followed by a second External Function instructing a specified peripheral to request data from the DOC. (This instruction consists of an RDUC - Receive Data (from) Unit Computer - command and a bit position encoded address.) The specified peripheral responds with an ODR and the DOC sends the data and and OA. Each data word sent in this fashion requires a separate ODR, and each data word will be sent with a separate OA.)

3.2.2.2.1 Operating modes. - The DOC interface shall have three modes of operation. These three modes will be Normal mode, External Function mode, and Output Acknowledge mode.

3.2.2.2.1.1 Normal mode. - The computer shall command the SDC to transfer data via the DOC interface, normal mode, using a format 1A Command Word having bits 9 and 10 clear. When the SDC receives a format 1A command with bits 9 and 10 clear, the DOC interface shall immediately and internally generate, and place on the DOC data lines a DOC-MSTC External Function word (see Figure 6) to master clear all

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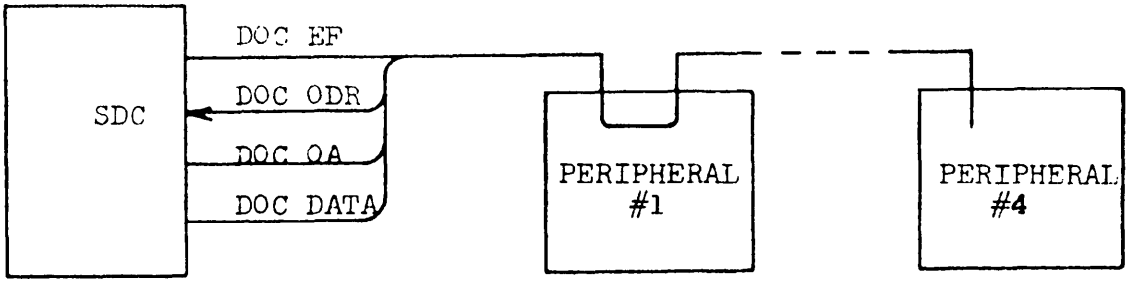


Figure 5. DOC Interface

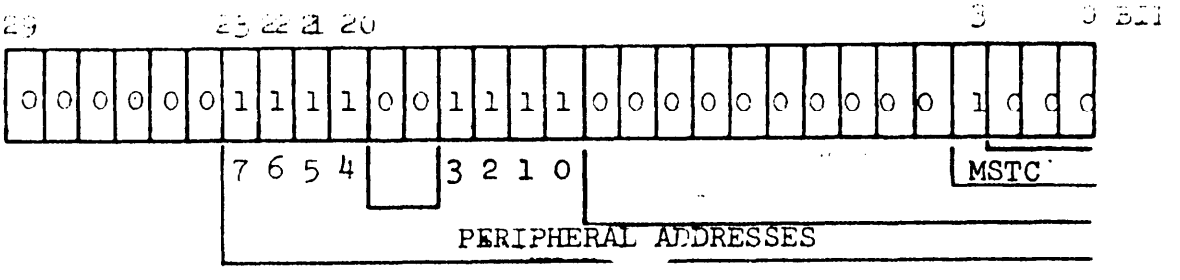


Figure 6. DOC - MSTC External Function Word

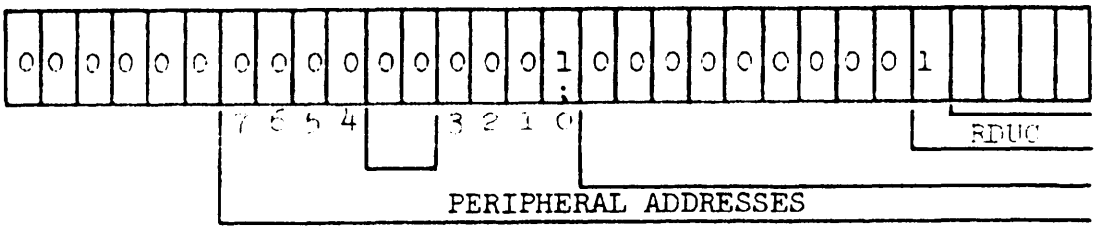


Figure 8. DOC - RDUC-C External Function Word

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the peripherals on the DOC. Accompanying the MSTC shall be an External Function pulse on the EF control line (see Figure 7). Following the MSTC word the DOC interface shall generate and place on the DOC data lines a DOC-RDUC External Function Word (see Figure 8) to the peripheral addressed by the computer. The RDUC word will be accompanied by an External Function pulse on the EF control line (see Figure 7).

After the second External Function word the data transmission sequence shall begin. The address peripheral will transition the ODR control line from logic 0 to logic 1. As soon as the SDC receives the data from the computer and the peripheral has raised the ODR, the data shall be sent accompanied by an Output Acknowledge pulse on the OA control line (see Figure 7). The ODR-OA sequence will continue until all the data words have been transmitted from the computer to the peripheral. If N words were to be transmitted, than N + 1 ODR transition from logic 0 to logic 1 shall not be acknowledged by the SDC.

3.2.2.2.1.2 External Function mode. - The computer shall cause the DOC interface in the SDC to enter the External Function mode by setting bit 10 in the Format 1A Command Word. In the External Function mode the DOC interface shall not automatically generate the MSTC or RDUC External Function words described in the preceding paragraph. Rather, the DOC interface shall place the data from the computer on the DOC data lines and shall place a pulse on the EF control line. The width of the pulse shall be 15 microseconds ±5 microseconds, measured at the 50 percent points. Each following data word from the computer to be transmitted to the External Function mode shall be transmitted with a similar External Function pulse.

3.2.2.2.1.3 Output Acknowledge mode. - The computer shall cause the DOC interface in the SDC to enter the Output Acknowledge mode by setting bit 9 in the format 1A Command Word. In the Output Acknowledge mode the DOC interface shall not automatically generate the MSTC or RDUC External Function words.

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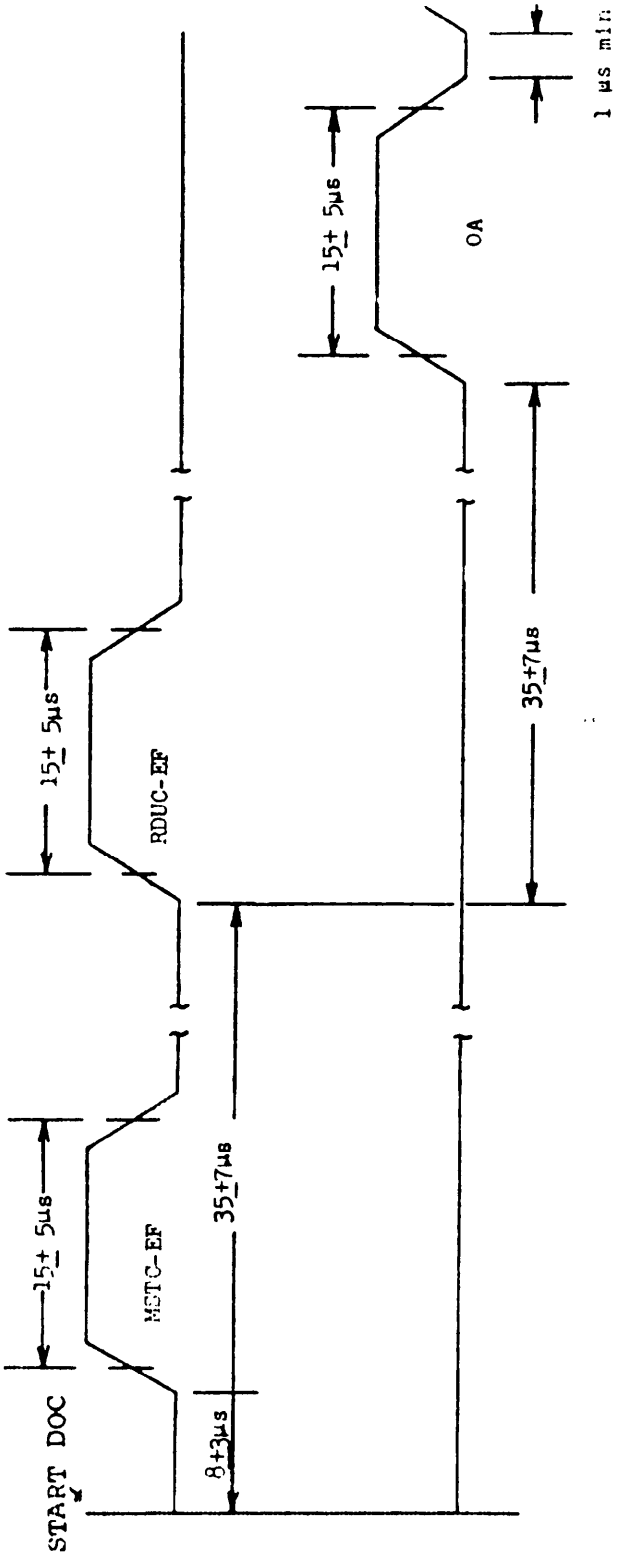


Figure 7. DOC Normal Mode Timing

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To transmit data in the Output Acknowledge mode the peripheral must first request the data by placing the ODR line in the logic 1 state. As soon as the data has been received from the computer and placed upon the data lines, and ODR has been received, the DOC interface shall place an Output Acknowledge pulse on the OA control line. The width of the pulse shall be 15 microseconds ± 5 microseconds measured at the 50 percent point. Each following data word from the computer to be transmitted in the Output Acknowledge mode shall require a request (ODR) from the peripheral before the Output Acknowledge pulse will be sent.

3.2.2.2.2 Buffer limits and timing considerations. - The DOC interface shall have the capability to transfer from 1 to 511 continuous data words to any one peripheral. The number of continuous data words is contained in binary format in the first nine bits of format 1A. A one word transfer is indicated by either a one or a zero in bit zero of the format 1A Command Word. For data buffers of more than one word, the SDC shall continue to ask the computer for the next data word as soon as the previous word has been transmitted on the DOC, until the indicated buffer has been completed.

On data transfers where the DOC interface is required to wait for an Output Data Request, a maximum wait time of 100 ± 15 microseconds shall be allowed. If the ODR is not received before the termination of the timed wait period, a DOC time-out fault shall be generated. This time out fault shall force an Output Acknowledge pulse to be generated. The DOC time out fault shall cause bit 27 in the SDC fault register to be set.

The occurence of a time out fault shall not terminate the DOC data buffer.

3.2.2.2.3 Level shifters - line drivers and receivers. - One set of 30 output data line drivers shall be used in parallel to drive all the peripheral devices on one DOC. The SDC shall have separate line drivers for the transmission of control signals (External Function and Output Acknowledge) and shall have separate input amplifiers for the receipt of control signals (Output Data Request).

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The data line drivers shall have the capability of driving at least 300 feet of standard Navy type 2A-40 cable. Each line driver shall be able to deliver at least 50 milliamperes from a source impedance of not more than 25 ohms.

3.2.2.3 Logic input channel (LIC) interface. - The LIC interface circuits must allow communication with all devices connected to the LIC's. Each LIC contains 30 bits of relay switch closures. Provisions shall be made for processing LIC bits which are controlled by three different relay voltages: +28 volts dc, +50 volts dc, and 115 volts ac, 60 hertz or 400 hertz.

A reference supply received from the external equipment shall be connected to one side of the relay's coil (several relays may use the same supply). The other side of the coil shall be connected to a line which carries the specific status signal associated with that relay. When the status signal is active, it shall cause the relay to be activated. Suppression shall be performed on the input to the relay coils. A peripheral device connected to the LIC shall either send a relay energizing signal to indicate a logic "1", or it shall not send the signal to indicate a logic "0". The SDC shall then transfer the logic information to the controlling computer. Each LIC shall be assigned a channel address and is selected by control word format 1.

The LIC shall have the capability of internally bussing common return wires together by the use of a specially designed PC Card. The bussing format as well as the relay control voltages shall be specified by the purchaser at a time of order.

3.2.2.4 Logic output channel (LOC) interface. - The LOC interface circuits must allow communication with all devices connected to the LOC channels. Each LOC channel contains 30 bits of logic information. They must be capable of passing the following currents at the voltages given through any LOC channel relay contact: 0.5 amperes at 50 volts dc, and 1.0 amperes at 115 volts ac, 60 hertz or 400 hertz. An output voltage represents a logic one. Each LOC shall be assigned a channel address. The controlling computer

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shall address a LOC with the control word, format 1. Upon recognizing the LOC address, the SDC raises an ODR to the computer which sends 30 bits of LOC data, format 2.

3.2.3 Interface requirements for analog data. - The SDC shall have channel capacities as shown in Table III.

3.2.3.1 Signal conditioning. - The SDC shall use input signal conditioners to accept the various types of input analog signals (ac, dc, and synchro). Output signal conditioners will scale the analog output to ac, dc, or synchro output voltage levels.

3.2.3.1.1 DC input signal conditioning. - The dc input signal conditioners shall accept dc, analog input signals with references of ±80 volts dc or less. Reference tolerances shall have a differential input stage with minimum ground isolation of 100 kilohms and a minimum input impedance of 100 kilohms. The scaling of the input signal shall be effected by external connections to the signal conditioner. The 3 decibel bandwidth of a dc signal conditioner shall not be less than 1000 hertz. A two-wire input with shield shall be used for each dc input and reference voltage.

3.2.3.1.2 AC input signal conditioning. - The ac input signal conditioner shall scale ac analog input signals having references of ±24 volts rms or ±12 volts rms, 400 hertz. Both types of references must be within a ±5 percent voltage tolerance and within a ±5 percent frequency tolerance. The ac input signal conditioner shall have a differential input stage with a minimum input impedance of 100 kilohms and a minimum ground isolation of 100 kilohms. The input signal shall be scaled by external connections to the input signal conditioner. The 6 decibel bandwidth of these signal conditioners shall not be less than 40 hertz.

3.2.3.1.3 Synchro input signal. - The synchro input signal conditioner shall accept three-wire synchro input (with shield) signal 90 volts rms, 400 hertz, with a two-wire

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TABLE III - SDC channel capacity

	MOD 0 DDG 2/15	MOD 1 FT-D	MOD 2 TALOS	MOD 3 DLGN-38	MOD 4 DDG-31
A/D DC	8	8	3	7	8
A/D AC	4	3	3	3	8
S/D 1x	1	1	1	1	2
S/D 2x	9	15	7	9	11
D/S 1x	2	1	0	1	2
D/S 2x	2	2	1	0	2
D/A DC	19	13	5	12	23
D/A AC	15	2	2	2	20
DOC	2	1	1	1	2
DD/DDI	2	2	1	1	2
LOC	2	3	2	3	2
LIC	2	2	1	2	2

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(with shield) 115 volts rms phase reference. The 3 decibel bandwidth of the signal conditioner shall not be less than 40 hertz. Shipboard voltage and frequency variations shall prevail on all synchro signals and references. The synchro input signal conditioner shall have a minimum input impedance of 100 kilohms and a minimum ground isolation of 10 megohms.

The SDC shall have the capability of accepting 60 hertz synchro inputs with a degraded accuracy, by means of resistive padding the input signal down to a level that can be accepted by the standard 400 hertz synchro input channel. The converter shall have an accuracy of ±2 degrees when converting these 60 hertz input synchro signals.

3.2.3.1.4 DC output signal conditioning. - The SDC shall provide for a maximum full scale dc output signal level of ±80 volts dc at ±8 milliamperes, maximum, or ±20 volts dc at ±40 milliamperes, maximum, but not both on a specified channel. The signal conditioners shall have a maximum output impedance at the SDC's output of less than 5 ohms and a minimum ground isolation of 100 kilohms. The output signal conditioner shall not be impared by a short circuit of indefinite duration at its output. It shall recover, to operate as specified, after removal of the short circuit. Any ac ripple on the dc output signal shall not cause the output to fall below its specified accuracy.

3.2.3.1.5 AC output signal conditioning. - The SDC shall provide for the full scale output analog voltage level of either 12 volts rms at 25 milliamperes rms or 24 volts rms at 3.5 milliamperes rms and at frequencies up to 400 hertz. The signal conditioners shall have a minimum ground isolation of 100 kilohms. The maximum quadrature due to the SDC at 400 hertz at the load shall be 2.0 percent.

The output signal conditioner shall not be impaired by a short circuit on its output. It shall recover, to operate as specified, after removal of the short circuit.

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3.2.3.1.6 Synchro output signal conditioning. - The SDC will provide for a three-wire synchro signal 90 volts rms, ±1 percent 400 hertz at 2500 ohm line-to-line load, or 11.8 volts rms at 300 ohm line-to-line load, but not both on the specified channel. The synchro output signal conditioners shall have a minimum ground isolation of 10 megohms.

3.2.4 Analog conversion requirements. -

3.2.4.1 Linear signal A/D conversion. - Linear signal (ac or dc) A/D conversion shall be performed with no sample-and-hold circuitry. The A/D conversion shall be made in the SDC as a ratio of the input signal to either an external (ac or dc) or internal (dc) reference, depending upon the required accuracy of the selected channel. The accuracy of the A/D conversion shall be as shown in Table I.

After the A/D conversion, data shall be transferred in ones complement format to the controlling computer. The digital format of the binary equivalent of the analog input signal as it will appear to the computer is shown in Figure 9a.

3.2.4.2 Synchro-to-digital (S/D) conversion. -

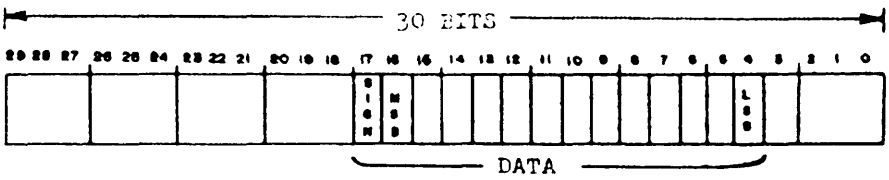
3.2.4.2.1 Single-speed synchro-to-digital (1x S/D) conversion. - The SDC shall perform single-speed S/D conversion with no sample-and-hold circuitry. After the SDC performs the 1x S/D conversion, data, in the standard binary angular measure (bam), shall be transferred to the controlling computer. The digital format of the converted one-speed synchro signal is shown in Figure 9b.

The accuracy of the conversion shall be as shown in Table I.

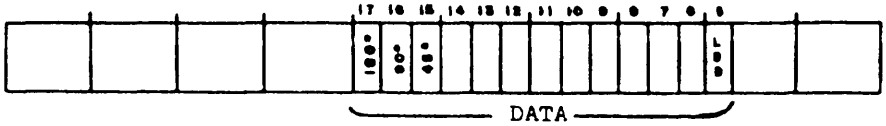
The maximum time required for single-speed S/D conversion shall be 150 microseconds, including multiplex switching.

3.2.4.2.2 Dual-speed synchro-to-digital (2x S/D) conversion. - The SDC shall perform two-speed S/D conversion without using any sample-and-hold circuitry. The fully merged result of the conversion in the standard "bam" format, shall be transferred to the controlling computer. The digital format of the converted 2x synchro signal is shown in Figure 9c.

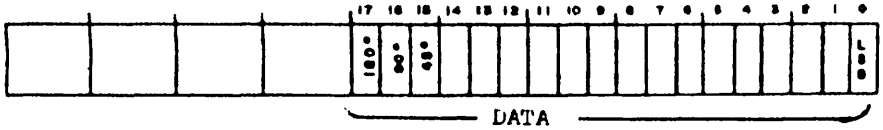
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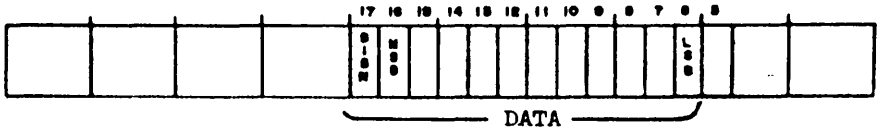
a) LINEAR A/D CONVERSION



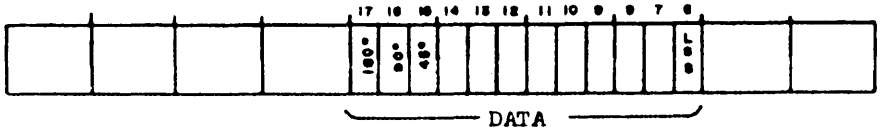
b) SYNCHRO TO DIGITAL CONVERSION ("BAM" FORMAT), 1-SPEED



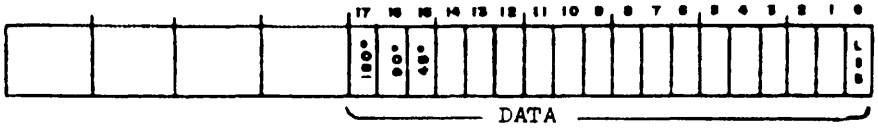
c) SYNCHRO TO DIGITAL CONVERSION, 2-SPEED



d) LINEAR D/A CONVERSION



e) DIGITAL-TO-SYNCHRO, 1-SPEED



f) DIGITAL-TO-SYNCHRO, 2-SPEED

Figure 9. Digital Formats of Analog Data

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The accuracy of the conversion shall be as shown in Table I.

3.2.4.3 Linear signal D/A conversion. - The SDC shall convert digital data from a computer to a linear ac or dc analog signal. The digital data received from the computer shall be in ones complement format as shown in Figure 9d.

The output deviation of a linear ac D/A converter shall not be more than 0.002 percent of full scale output owing to phase shift within the SDC. The harmonic distortion of the ac signal at the output shall be nominally equal to that of the supplied ac reference.

The accuracy of a linear ac D/A conversion shall be as shown in Table II. Resolution of the ac D/A conversion shall be 12 bits.

Two dc D/A conversion requirements shall be provided. The resolution shall be 10 bits for Type 1 and 12 bits for Type 2 respectively.

3.2.4.4 Digital/synchro (D/S) conversion. -

3.2.4.4.1 Digital/single-speed (D/1xs) conversion. - The SDC shall convert digital data from a computer to a single-speed 3-wire, 90 volts rms ±1 percent line-to-line, 400 hertz synchro signal. The digital data as transmitted by the computer shall be in standard "bam" format as shown in Figure 9e.

The total harmonic distortion of the 3-wire synchro signal, due to the SDC shall not exceed 2 percent. The accuracy of the D/1xs conversion shall be as shown in Table II.

3.2.4.4.2 Digital/dual-speed (D/2xs) conversion. - The SDC shall convert digital data from a computer to a dual-speed 3-wire, 90 volts rms ±1 percent, line-to-line 400 hertz synchro signal. The digital data received by the computer shall be in standard "bam" format as shown in Figure 9b.

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The total harmonic distortion of the dual-speed 3-wire synchro signal, due to the SDC shall not exceed 2 percent. The D/2xs conversion shall be accurate as shown in Table II.

3.2.4.5 Filtered outputs. - The SDC shall have two special 2X digital to synchro channels with filtered outputs. The filtering shall consist of a first order digital filter on each speed of the 2X data being transmitted to the synchros. The form of the digital filter shall be as shown in Figure 10.

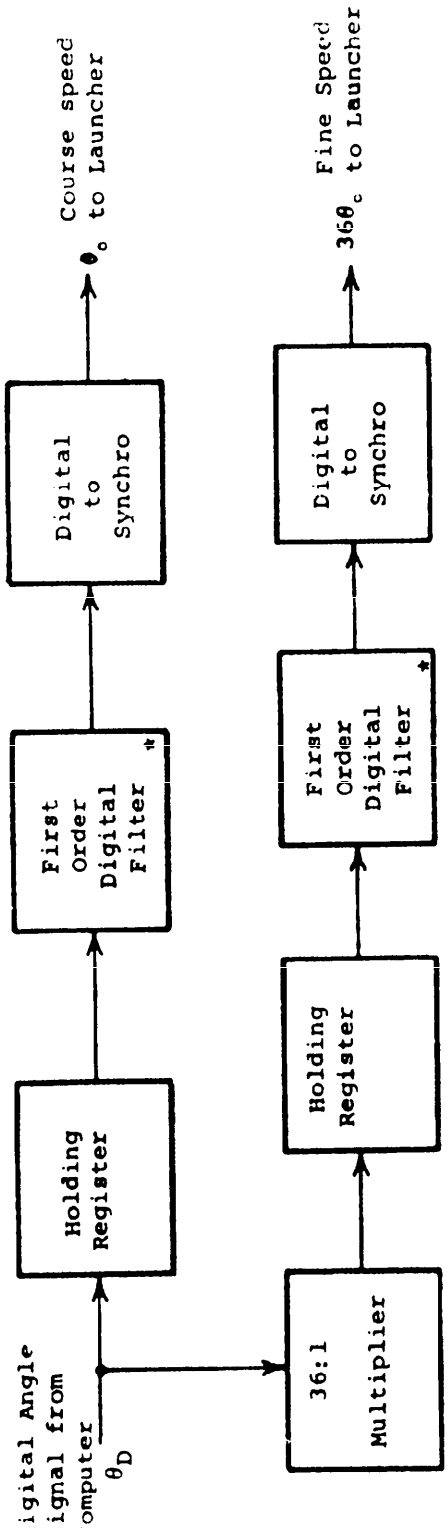
3.2.5 Computer interface control and timing. - Each input-output channel of the SDC shall function as specified herein. The SDC shall recognize each binary one received on any control line by detecting the change in voltage level of the signal. It shall not recognize another binary one on that control line until its circuitry has been reset by a change in the voltage level of the signal in the opposite direction (a binary zero).

3.2.5.1 Input communication. - When interconnected as shown in Figure 11, the SDC shall transfer data to the computer as specified herein. Input channel control lines shall function as specified in Table IV.

TABLE IV. - Function of input channel control line

Name of line	Direction of signal	Function
Input Data Request Line	SDC to computer	Set condition shall indicate that the SDC has placed a word of data available to the computer on the Input Data Lines of that channel.
Input Acknowledge Line	computer to SDC	Set condition will indicate that the computer has read the Input Data Lines of that channel.

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* This first order filter is an impulse invariant digital version of the analog filter

$$\frac{So}{S+So} \quad \text{with So} \approx 15\text{-}20 \text{ RAD/SEC}$$

The transfer function of the digital filter is

$$H(Z) = \frac{1/64}{1 - (63/64)Z^{-1}} \quad \text{with a 1 ms sampling interval}$$

Figure 10. Digital Figure

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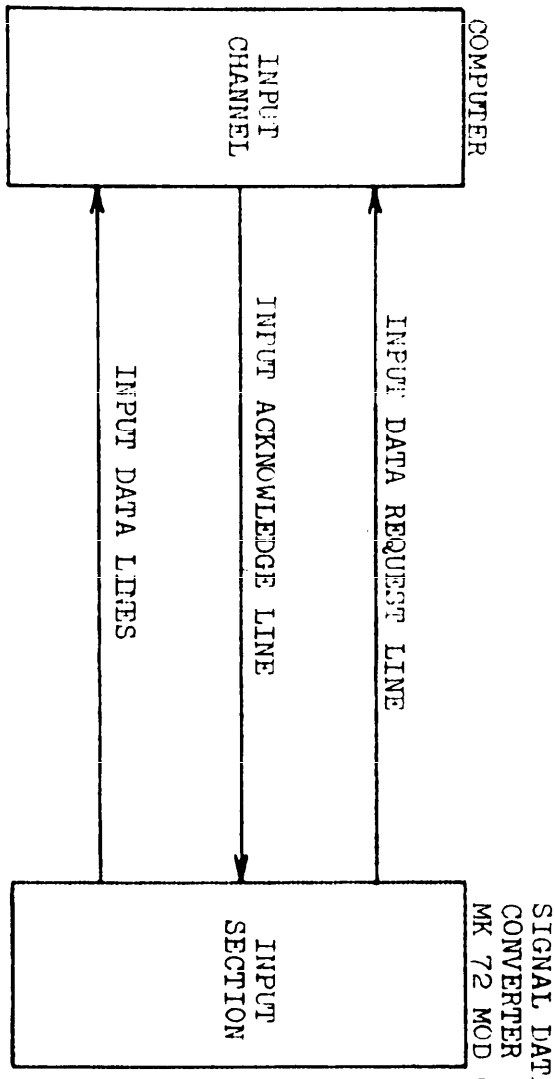


Figure 11. Input Communication Interface

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3.2.5.1.1 Transfer of input data. - Whenever an Input Data buffer has been established for a channel, the SDC shall transfer data as follows:

a) The SDC shall place a word of data on the Input Data Lines.
b) The SDC shall set the Input Data Request Line (to indicate that a word of data is on the Input Data Lines).
c) In accordance with internal priority, the computer will detect the setting of the Input Data Request Line.
d) The computer will read the data word which is on the Input Data Lines.
e) The computer will set the Input Acknowledge Line (indicating that it has read the data word on the Input Data Lines).
f) The SDC shall detect the setting of the Input Acknowledge Line. (The SDC may clear the Input Data Request Line any time after detecting the setting of the Input Acknowledge Line, but it shall clear the Input Data Request before the computer will recognize the next Input Data Request.)
g) The computer will clear the Input Acknowledge Line before it reads the next word on the Input Data Lines.

The SDC shall repeat this sequence for each successive word of data.

3.2.5.2 Output communication. - When interconnected as shown in Figure 12, the SDC shall accept data from the computer as specified herein. Output channel control lines shall function as specified in Table V.

TABLE V. - Function of output channel control line

Name of Line	Direction of signal	Function
External Function Acknowledge Line	computer to SDC	Set condition will indicate that the computer has placed an External Function Code Word on the Output Data Lines of that channel.

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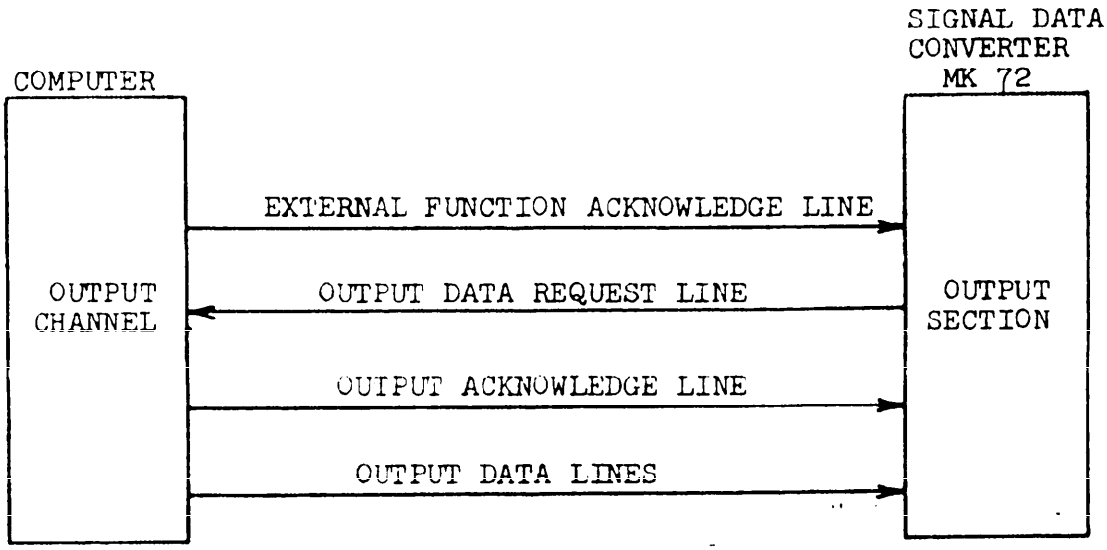


Figure 12. Output Communication Interface

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TABLE V. (continued)

Name of Line	Direction of signal	Function
Output Data Request Line	SDC to computer	Set condition will indicate readiness of the SDC to accept a word of data on that channel.
Output Acknowledge Line	computer to SDC	Set condition will indicate that the computer has placed a word of data on the Output Data Lines of that channel.

3.2.5.2.1 Transfer of output data. - Whenever an Output Data buffer has been established for a channel, the SDC shall accept data as follows:

a) When the SDC is ready to accept data, it shall set the Output Data Request Line. (This may already have happened before the Output Data buffer was established.)
b) In accordance with internal priority, the computer will detect the setting of the Output Data Request Line.
c) The computer will place a word of data on the Output Data Lines.
d) The computer will set the Output Acknowledge Line (to indicate that a word of data is on the Output Data Lines).
e) The SDC shall detect the setting of the Output Acknowledge Line. (The SDC may clear the Output Data Request Line any time after detecting the setting of the Output Acknowledge Line, but it shall clear the Output Data Request Line before the computer will recognize the next Output Data Request.)
f) The SDC shall read the data word which is on the Output Data Lines.
g) The computer will clear the Output Acknowledge Line before it places the next word on the Output Data Lines.

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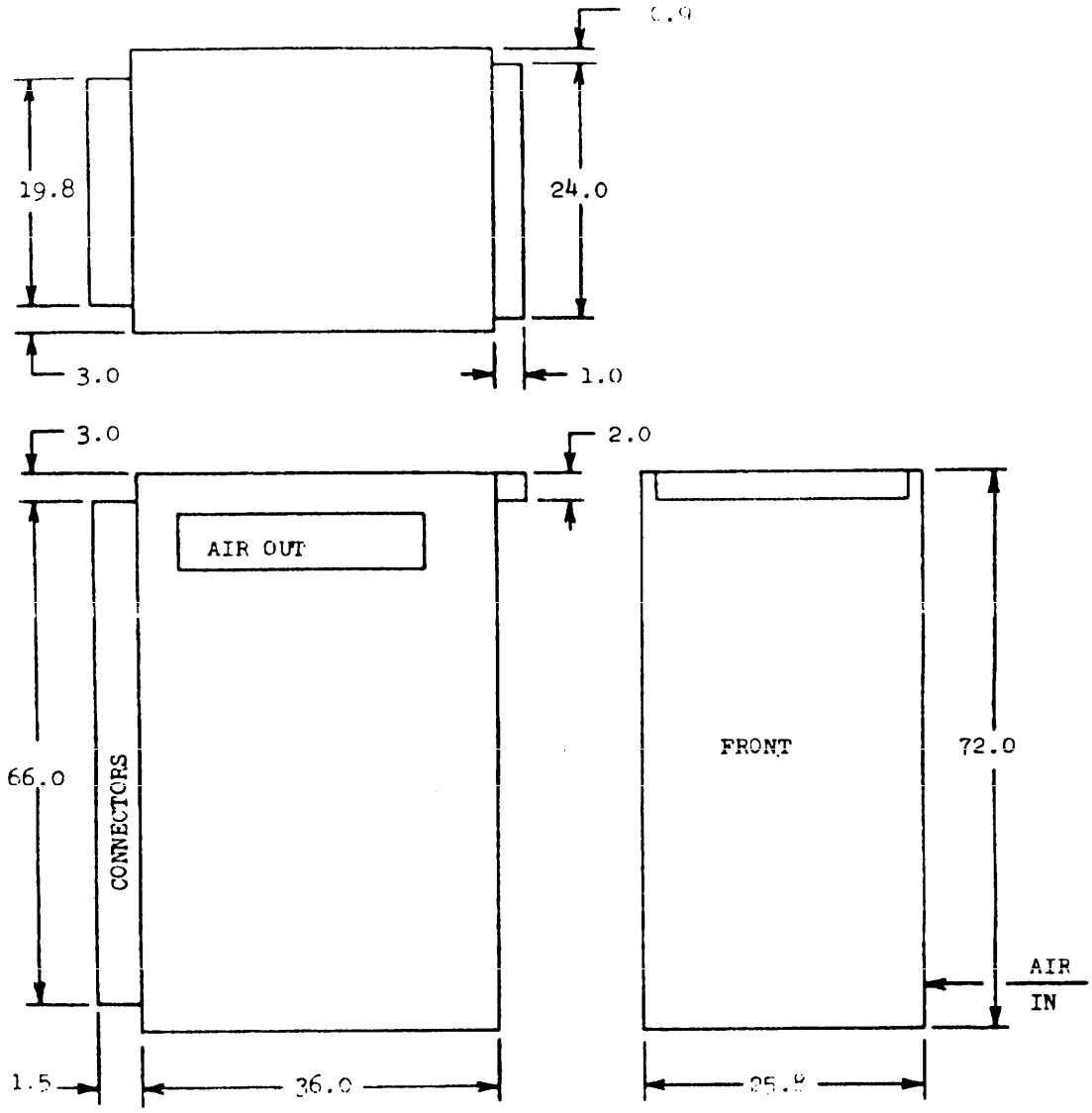
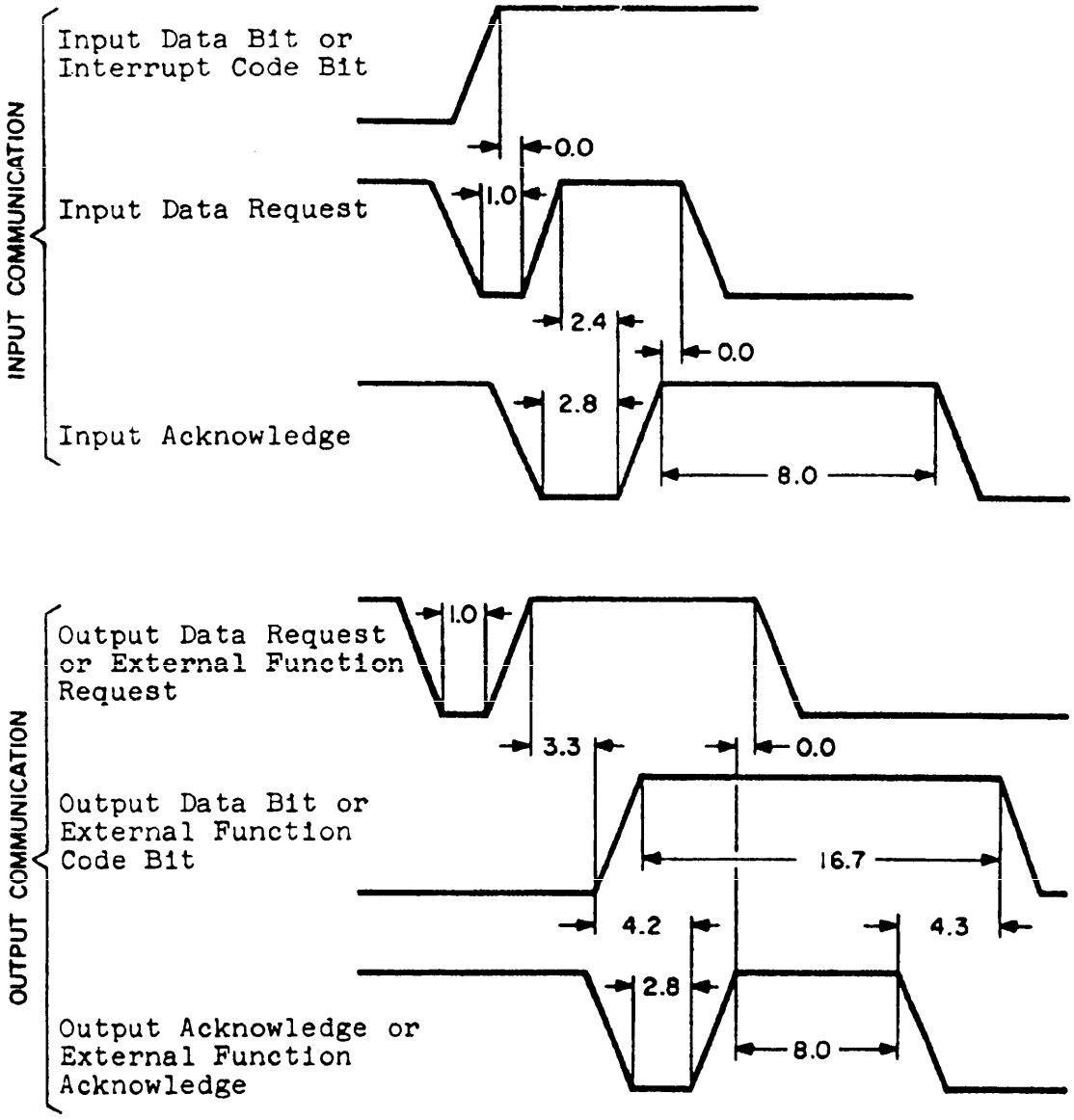


Figure 14. SDC-Cabinet Outline

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NOTES: Upper level of each signal is a binary one; lower level is a binary zero. All times are in microseconds, are minimums, and are measured between points on the signal pulse that are either 10 percent or 90 percent of the amplitude as shown. All transitions have a maximum duration of 6 microseconds and a maximum rate of 5 volts per microsecond.

Figure 13. Slow Interface Timing

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c) The output of the circuit shall not switch as a result of any input transient-pulse signal that has an integrated amplitude-duration of less than 15 volt-microseconds. (A delay of 1.5 ±0.5 microseconds with a 15-volt step input.)
d) The circuit shall not draw more than 4.0 milliamperes for a steady-state binary one nor more than 1.0 milliampere for a steady-state binary zero.
e) The output of the circuit shall be a binary zero whenever the steady state input signal is more negative than -7.5 volts.
f) The output of the circuit shall be a binary one whenever the steady-state input signal is more positive than -4.5 volts.

3.3.1.1.2 Line drivers. - Each slow interface line driver circuit shall have the following characteristics when driving a line with any capacitance of up to 6000 picofarads. Each data line driver circuit that drives more than one line shall have proportionally more capability than specified here.

a) The steady-state output voltage representing a binary one shall be -1.5 volts to +1.5 volts.
b) The steady-state output voltage representing a binary zero shall be -10.0 volts to -17.5 volts.
c) The voltage variation between all binary zero output signals on one channel shall not exceed 1.0 volt.
d) The voltage variation between all binary one output signals on one channel shall not exceed 1.0 volt.
e) The circuit shall be capable of supplying 4.0 milliamperes per line for a steady-state binary one output.
f) The circuit shall be capable of supplying 1.0 milliampere per line for a steady-state binary zero output.
g) The output of the circuit shall switch in not more than 6.0 microseconds (measured between 10 and 90 percent amplitude points).

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h) The output of the circuit shall switch at a rate of not more than 5.0 volts per microsecond. The
i) The circuit shall be capable of driving the line and its termination (input amplifier) while switching as well as in the steady state condition.
j) The driver circuit shall present not less than 100,000 ohms impedance to the cable whenever power to a control line driver circuit is removed.

3.3.1.2 Reference voltages. - The SDC shall provide two ±12 volts rms, 400 hertz reference voltages for external equipment. These references will be compensated for line drops due to loads by maintaining remote and separate ±12 volts rms reference response signals from the external equipments. The nominal value of these references shall be ±12 volts rms ±2 percent with an input 115 volts rms line regulation of 0.5 percent. The maximum load on each voltage line shall be 500 ohms resistive.

(All of the references specified above are derived from the 115 volts rms ±0.5 percent input line by means of a stepdown transformer.)

The SDC shall provide one +12 volts ac, 400 hertz reference voltage for external use. The reference shall be ±12 volts rms ±0.5 percent and have an accuracy of 0.5 percent due to input line regulation. The specified accuracy shall be delivered to a load of more than 20 kilohms, at the end of 300 feet of standard Navy cable.

3.3.1.3 External power requirements. - The SDC shall operate from the following sources of primary ship power:

1) 115 volts ac, 3-phase, 400 hertz, Type 3
2) 115 volts ac, 1-phase, 60 hertz, Type 1
3) 115 volts ac, 1-phase, 60 hertz, Ship Service
4) +50 volts dc, ±5 volts regulation
5) 115 volts ac, 1-phase, ±0.5 percent (for ac reference only)

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3.3.1.4 Cabling. - Cabling and wiring shall be in accordance with MIL-F-18870. Cable connectors shall be provided for attaching signal cables between the SDC and other equipments.

3.3.1.5 Grounding. - The enclosure for the SDC shall be grounded to a single common ground point within the ship's structure by means of a suitable ground strap. Circuit grounding shall be in accordance with MIL-F-18870.

3.3.2 Physical characteristics. -

3.3.2.1 Size. - The SDC shall conform to the maximum outline drawing in Figure 14.

3.3.2.2 Weight. - The SDC shall not weigh more than 1050 pounds.

3.3.2.3 Cooling interface. - The SDC shall be entirely cooled by air within the FCS compartment. It shall be capableof operating in a class 4 temperature environment as defined in MIL-F-18870.

3.3.2.4 Mechanical noise. -

3.3.2.4.1 Enclosed area airborne noise. - Equipment in enclosed areas shall not generate airborne noise in excess of that permitted for Grade C equipment in MIL-STD-740.

3.3.2.4.2 Structure-borne noise. - The equipment shall fall within the requirements of Type 3 in accordance with MIL-STD-740.

3.3.2.5 Safety. - Safety (personnel hazards) requirements for the SDC shall be in accordance with requirement 1 of MIL-STD-454.

3.4 Electromagnetic Compatibility (EMC). - Electromagnetic interference characteristics of the SDC (emission and susceptibility) shall be in accordance with the test requirement of Table II for Equipment Class IC as set forth in MIL-STD-461. Test Requirement (T)(CS07) shall not apply.

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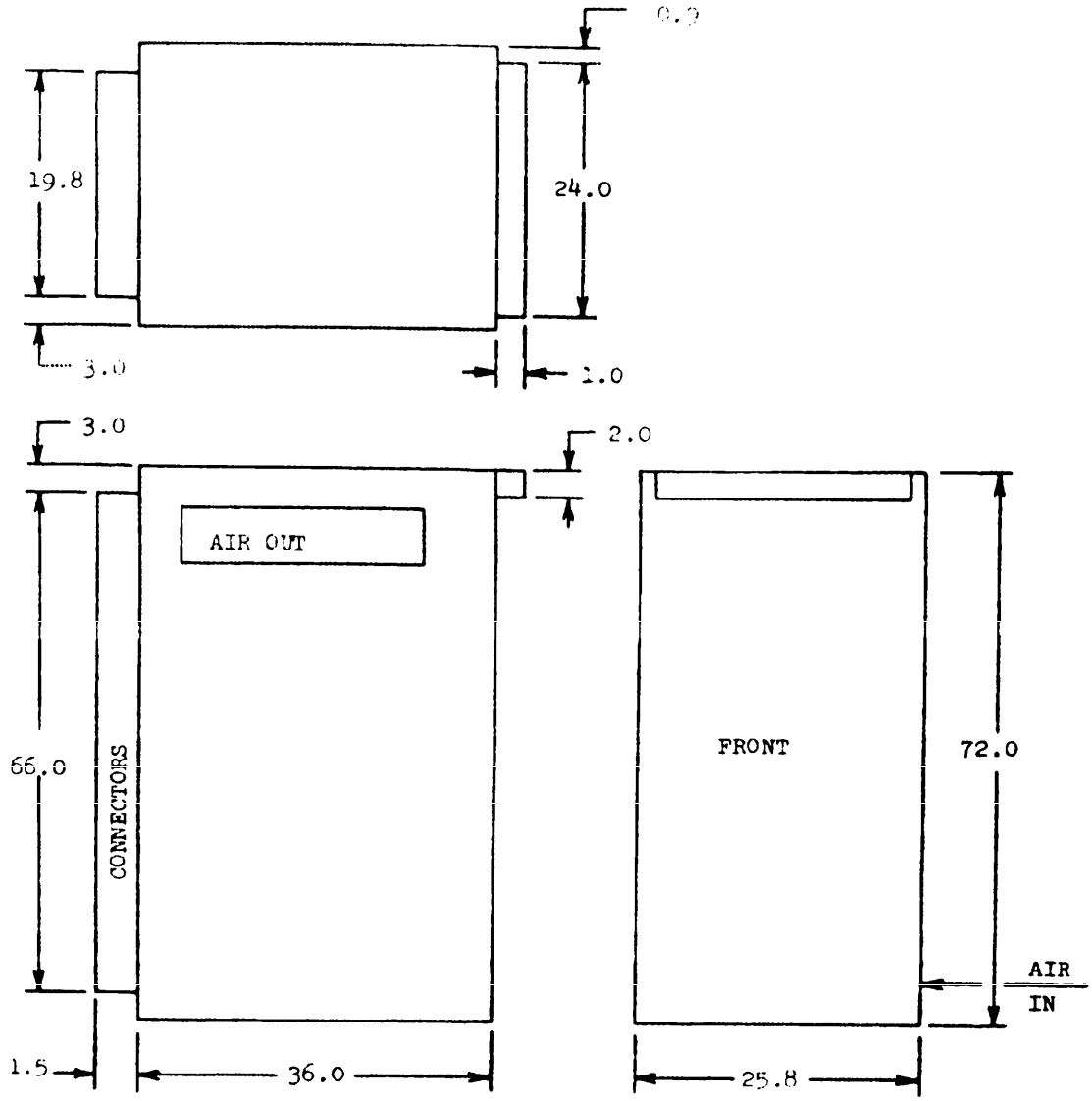


Figure 14. SDC-Cabinet Outline

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3.5 Environmental requirements. - The SDC shall be capable of continuously operating in the presence of shipboard world-wide environment as defined by MIL-STD-210, except as noted herein. No damage to the SDC shall occur when subjected to short term storage and transit as defined in MIL-STD-210.

3.5.1 Centrifugal force. - The SDC shall, without degradation of performance, withstand a centrifugal force, experienced during a full maneuver turn of the vessel, equal to one-tenth of the weight of the equipment unit under test.

3.5.2 Ship roll and pitch. - The SDC shall operate without degradation of performance under conditions of roll from 30 degrees port to 30 degrees starboard within a period of 9 seconds, and pitch from 5 degrees up to 5 degrees down within a period of 6 seconds.

3.5.3 Shock. - The SDC shall withstand shock for Grade A, hull mounted, principal unit, Class I, medium weight equipment as set forth in MIL-S-901.

3.5.4 Vibration. - The SDC shall meet the conditions of vibration and acceleration as specified under Type I environment in MIL-STD-167, except that the upper frequency limit shall be 25 hertz.

3.5.5 Humidity and ambient temperature. - The SDC shall start and operate as specified herein under relative humidity conditions not exceeding 95 percent, and under ambient temperature ranges of from 0 degrees to 50 degrees C (32 degrees to 122 degrees F).

3.5.6 Corrosion, moisture and fungus. - The selection of moisture and fungus-proofing treatment, fungicidal agents, and corrosion and fungus resistant meterials, finishes, and coatings shall be in accordance with MIL-F-18870.

3.6 Reliability. - Each SDC shall have a mean-time-between-failures (MTBF) of not less than 1000 hours.

3.7 Maintainability. - The mean-time-to-repair (MTTR) of the SDC shall not exceed 30 minutes.

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3.7.1 Interchangeability. - All parts shall be interchangeable physically and electrically with corresponding parts without drilling, filing, or the use of undue force. The replacement part when installed shall not result in a departure from the performance specified herein. The SDC itself shall be interchangeable with the same type of the SDC.

3.8 Government loaned equipment. - Government loaned equipment shall be as specified in the contract or order.

3.9 Government furnished equipment. - Government furnished equipment shall be as specified in the contract or order.

3.10 Workmanship. - Workmanship shall be in accordance with Requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. - Each SDC offered for acceptance under contract shall be subjected to the examinations and tests set forth in QAP 2868644 or QAP 2884550 and as may otherwise be specified herein. In addition, samples selected from units having satisfactorily passed these inspections will, as specified in the contract or order, be subjected to the tests of 4.2.3 through 4.2.6 at an activity designated by the contracting officer. Sampling for these tests shall be as determined by the contract or order. Documentation describing paramerers and procedures for these inspections shall be as specified in the

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contract or order. Failure to meet any of the inspection requirements shall be considered cause for rejection. Acceptance of production units will be based on satisfactory performance of the samples to the test as required by the contract or order.

4.2.1 Power system variations. - The SDC shall be tested for satisfactory operation under the power system variations characterizing the power sources specified herein. This test may be conducted during the performance of any or all of the tests of this specification at the discretion of the testing activity or as specified by the contract or order.

4.2.2 Environmental tests. - The SDC shall be subject to environmental tests specified herein. Failure to meet these environmental requirements shall be considered cause for rejection.

4.2.2.1 Temperature. - The SDC shall be subjected to the operating and non-operating temperature tests as specified for class-4 equipments in MIL-F-18870.

4.2.2.2 Humidity. - Humidity tests shall be conducted in accordance with MIL-F-18870.

4.2.2.3 Shock and vibration. - The SDC shall be subjected to the Type A, medium weight shock tests for Grade A, Class I equipment as specified in MIL-S-901 and the Type I vibration tests specified in MIL-STD-167. The tests shall be conducted with the equipment secured to the mounting brackets of the test machine in the same manner that it will be secured on shipboard. The equipment shall be energized and be in operating condition during these tests.

4.2.2.4 Mechanical noise tests. - Mechanical noise tests shall be conducted in accordance with MIL-STD-740 to determine conformance with the requirements of this specification.

4.2.2.5 Salt spray tests. - A salt spray test in accordance with FED-STD-151A test Method 811 shall be conducted

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on all parts with coatings or finishes, and to frames and enclosure structures assembled for final use. The test shall not be conducted on complete sets or major units. Sample corner structures and other critical sections shall be used for the test. Duration of the tests shall be 200 hours for frame and enclosure structures and 48 hours for individual parts.

4.2.2.6 Inclination test. - The SDC shall be subjected to the inclination test set forth in MIL-F-18870.

4.2.3 EMC tests. - The EMC requirements for the SDC shall be tested to verify compliance with MIL-STD-461 using test procedures set forth in MIL-STD-462.

4.2.4 Reliability. - When specified in the contract or order, reliability verification shall be in accordance with MIL-STD-781, Test Plan D, Level A, except that vibration shall not be performed.

4.2.4.1 Sources of failure data. - MTBF predictions shall be based on failure rate data from MIL-HDBK-217. For parts not covered in MIL-HDBK-217 existing data shall be used. The source of the existing data shall be specified and substantiated.

4.2.5 Maintainability. - When specified in the contract or order, maintainability verification shall be in accordance with MIL-STD-471.

4.2.5.1 Sources of maintainability data. - Maintainability predictions shall be based on data given in MIL-HDBK-472.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging of the SDC shall be specified in the contract or order.

5.2 Packing. - The SDC shall be packed as set forth in the contract or order.

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5.3 Marking. - Shipping containers shall be marked in accordance with MIL-STD-129, including serial numbers.

6. NOTES

6.1 Intended use. - The equipment covered by this specification, when installed aboard ship and integrated with other equipments, is intended for use in the fire control of TARTAR guided missiles and, if required, gun control.

6.2 Ordering data. - Procurement documents should include the following:

a) Title, identification number and date of this specification.
b) Documentation required, if any (see 3.7).
c) Government loaned equipment (see 3.8).
d) Government furnished equipment (see 3.9).
e) Number of equipment to the subjected to the inspections of 4.2.3 through 4.2.6 (see 4.2).
f) Verification of reliability requirements (see 4.2.5).
g) Verification of maintainability requirements (see 4.2.6).
h) Requirements for Preparation for delivery (see section 5).
i) Power system variation test requirements (see 4.2.2).

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